A positive contact test between Nipissing diabase and Gowganda argillites[1]

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A positive contact test between Gowganda argillites and Nipissing diabase indicates the shallowly inclined and southerly directed Nipissing remanence records a primary intrusive event. Contact tests on the steeply inclined Nipissing direction are also positive. Together these tests conclusively indicate there were two distinct phases of Nipissing intrusion. Limited evidence suggests that the sequence between these two events should be from the shallow to the steeply inclined direction, giving a pole path from south to north. This is in direct conflict with the presently accepted north to south trending Track 5.

Un test de contact positif entre les argillites de Gowganda et la diabase de Nipissing indique que la rémanence de Nipissing, faiblement inclinée et dirigée vers le sud, enregistre un épisode intrusif primaire. Les tests de contact sur la direction fortement inclinée de Nipissing sont aussi positifs. L'ensemble de ces tests indique de façon concluante qu'il y a eu deux phases distinctes dans l'intrusion de Nipissing. Des preuves moins concluantes suggèrent que la séquence chronologique entre les deux épisodes devrait être de la direction peu inclinée vers la direction à forte inclinaison, ce qui suppose un cheminement polaire du sud au nord. Ceci est en contradiction avec le cheminement généralement accepté de la Trace 5 qui va du nord au sud.

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Introduction

Two distinct remanence directions (N1 and N2) have been reported by many authors from the Nipissing diabase and Abitibi dykes (for example, Larochelle 1966; Symons 1967, 1970, 1971; Patel 1971; Symons and Londry 1975; Roy and Lapointe 1976; Irving and Naldrett 1977). The more commonly observed N1 direction has been found with both normal and reversed polarities. The average inclination of this direction (N1) is approximately 40°, however, the declination varies from 165° (Patel and Palmer 1974) to 209° (Symons 1967). The more southerly directed remanences usually have reversed polarity, while the more westerly directed remanences have normal polarity. This suggests that the N1 directions were acquired over a distinct interval of time. In contrast, the steeply inclined, N2, direction ($D = 265°$, $I = +70°$) is found with only normal polarity, and rarely shows much variation of remanence direction, suggesting that the magnetization records a single event. On the basis of well-executed contact tests it has been shown that the N2 direction, both in the Nipissing (Roy and Lapointe 1976) and the Abitibi (Irving and Naldrett 1977), records a primary remanence that originates from the time of intrusion. If this direction, N2, is primary, what mechanism then gave rise to the direction N1? The logical explanation for two distinct primary pole positions is two temporally distinct phases of Nipissing–Abitibi intrusion as originally proposed by Symons (1971). Later, however, Symons and Londry (1975) rejected the idea of two intrusive events, because they recognized what they considered to be both N1 and N2 remanence directions in the *same* Nipissing sill. It is the purpose of this note to report a positive contact test between Gowganda argillites and Nipissing diabase, both of which carry the normal polarity N1 remanence direction, and to discuss the evidence for two phases of intrusion, and their sequence of formation.

If then there is paleomagnetic evidence for more than one intrusive event, is this reflected in the radiometric data? A number of age dates have been reported from both the Abitibi and Nipissing intrusives. Gates and Hurley (1973) reported a Rb/Sr isochron of 2147 ± 68 Ma on ENEW Abitibi dikes carrying the N2 direction. Somewhat similar ages of 2155 ± 80 Ma by Van Schmus (1965) and of 2162 ± 27 Ma by Fairbairn *et al.* (1969) have been reported, both from Nipissing diabase sills carrying the N1 direction. In a recent $^{40}Ar/^{39}Ar$ study of Abitibi intrusives, Hanes and York (1978) found evidence for three distinct isotopic events at approximately 2150, 1800, and 1400 Ma. The contact test of Irving and Naldrett (1977), combined with the $^{40}Ar/^{39}Ar$ work of Hanes and York (1978), suggests the 2150 Ma age applies to the N2 remanence

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direction. As presently interpreted, pole path and magnetic stability data are taken as indicating that N1 postdates N2, with N1 having an age of approximately 1800 Ma (Roy and Lapointe 1976; Irving and McGlynn 1976). By no means, however, is this evidence conclusive, though it does agree with the second isotopic event recognized by Hanes and York (1978). This particular interpretation of the intrusive sequence as N2 → N1 has not always been in vogue. Symons (1971) recognizing that Nipissing sills carried N1, whereas the crosscutting dikes carried N2, suggested that the sequence should be N1 → N2. On this interpretation the dikes would be slightly younger than the sills. Because of inherent errors in geochronological methods, it is possible that this age difference could go unrecognized. For example, errors in the presently available radiometric data permit an age difference between the dikes and sills of at least 75 Ma, a time difference that would be apparent in paleomagnetic data. Hence, at present, the sequence of intrusion and the age of the N2 intrusive event remain uncertain.

The Contact Test

The test is based on five sites (60 specimens) from the centre of the Quirke Lake syncline just north of Elliot Lake (Twp. 150, 46°29′N; 82°38′W). Two sites were occupied in Nipissing diabase, one in the coarse-grained central portion of the intrusive (site 1), and one from the finer-grained chilled marginal phase (site 2). Three sites (3, 4, 5) were occupied in approximately the same stratigraphic horizon in the adjacent Gowganda argillites. Sites 3 and 4 are in the baked zone, while site 5 appears unaffected by the intrusion (Fig. 1). Data from site 5, which have been previously reported in a regional study of Gowganda and Chibougamau argillites (Morris 1977), are similar to results from other sites occupied in that study. Their remanence is dominated by a regional overprint post-folding magnetization which had a mean direction of $D = 338°$, $I = +68°$ yielding a pole at 184°E; 74°N.

Upon thermal and alternating field demagnetization, specimens from both sites 3 and 4 exhibited very limited direction change. In all cases the remanence appears to be univectorial ($D = 225°$, $I = +45°$, $\alpha_{95} = 5°$). The site mean directions calculated on alternating field and thermally cleaned specimens are statistically identical. The intensity decay characteristics suggest that the dominant remanence carrier is magnetite. Over 95% of the remanence is removed by 30 mT and 580°C. Comparison of baked (sites 3, 4) and unbaked (site 5) specimens

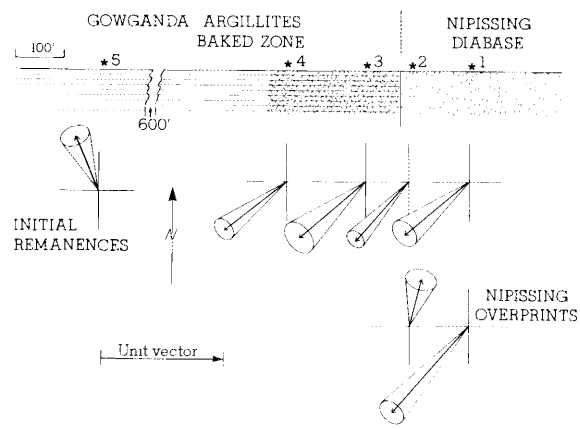


FIG. 1. Above: the distribution of sampling localities. Sites 1 and 2 are in the Nipissing diabase, sites 3 and 4 are in baked argillites, and site 5 represents the unbaked country rock. Below: the arrows represent the projection on the horizontal plane of a unit vector in the direction of the mean initial remanence direction at each site. True north is at the top of the diagram. All inclinations are positive; the length of the arrow is drawn proportional to the cosine of the site mean remanence inclination. The ellipses represent the 95% confidence limit in declination and inclination. The agreement between directions in the baked contact zone and Nipissing provides a positive contact test. Bottom right: shows the mean overprint directions found in the Nipissing diabase using the same convention as above.

show clear distinctions. The directions of remanence are statistically different (Fig. 1). In the baked specimens the coercivity spectrum is much lower and the Koenigsberger ratio much higher. There are also petrological differences. Examination of polished thin sections of samples from the baked zone shows that euhedral magnetites have been developed. These are more typical of samples from the higher regional metamorphic grades of Whitefish Falls (Morris 1977). Thin sections from site 5 show only secondary grain margin overgrowths, with no new secondary grains.

In contrast to the apparently simple magnetic history of the argillites, the two sites collected from the Nipissing diabase have extremely complex remanences. Probably in response to varying post-intrusion physico-chemical alteration processes, the coarse-grained specimens of site 1 have retained a magnetic record different to that retained by the fine-grained specimens of site 2. The natural remanent magnetization (NRM) directions of the coarse-grained phase are closely grouped around $D = 220°$, $I = +15°$ (OV1). Upon thermal demagnetization up to 550°C, the majority of specimens, whilst exhibiting very little directional change, show irregular oscillations of intensity. When these

data are plotted on vector diagrams (Roy and Park 1974) it can readily be seen that the NRM direction is mostly a composite of a dual-polarity remanence with a mean direction, irrespective of polarity, at $D = 222°, I = +19°$ ($\alpha_{95} = 7°$). A plot of the subtracted vectors (Fig. 2) shows that successive thermal demagnetization steps remove oppositely directed components with overlapping blocking temperature spectra. Upon further thermal treatment above 550°C, most specimens do not carry any meaningful remanence. A few specimens, however, do show a consistent directional shift towards a steeper inclination. The mean direction ($D = 231°, I = +49°, \alpha_{95} = 10°$) of this remanence is statistically identical (Fig. 1) to that observed in the baked argillites ($D = 225°, I = +45°, \alpha_{95} = 5°$) and therefore represents the initial magnetization of the intrusive.

The NRM of the fine-grained phase, which has a low mean remanent intensity of approximately 6 mA m^{-1}, has a steep positive inclination. Thermal demagnetization beyond 200°C generally removes this steep vector ($D = 014°, I = +67°$, OV2), and reveals another vector inclined at approximately 50° and directed towards the southwest. Further thermal treatment reveals this southwesterly remanence is again a composite vector of opposing polarities. Just as in the coarse-grained specimens, only with a steeper inclination, sequential demagnetization steps remove opposing polarities of the same remanence direction (Fig. 2). Again, we observe that the majority of the remanence resides in two remanences of opposing polarity with overlapping unblocking temperature spectra. The mean direction ($D = 223°, I = +52°, \alpha_{95} = 7°$) of this vector is statistically identical to the higher temperature direction observed in the coarse-grained diabase ($D = 231°, I = +49°$), and also to the direction observed in the baked argillites ($D = 225°, I = +45°$). Hence, as shown in Fig. 1, this evidence provides a conclusive and fully-executed positive contact test for the southerly Nipissing–Abitibi pole.

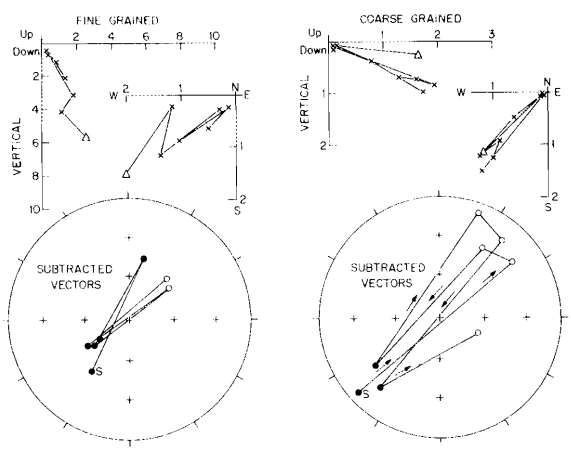


FIG. 2. Above: vector diagrams showing the sequential changes of direction and intensity observed in a typical specimen of, respectively, the fine- and coarse-grained Nipissing diabase during thermal demagnetization of 200, 300, 400, 500, 575, and 600°C. Open triangle represents the NRM direction; crosses represent the sequential demagnetization steps. Below: a stereonet plot of the sequential vectors removed during demagnetization. S represents the vector (NRM $- D, I_{200}$), followed by $D, I_{200} - D, I_{300}$ etc. Open (closed) dots are negatively (positively) inclined subtracted remanence vectors. Reversed and normal components of the same direction are removed by sequential demagnetization steps.

The complexity of the magnetizations observed in the Nipissing diabase, however, raises some important questions about the mode of acquisition of the various remanence components, and their resistance to subsequent remanence resetting. For example, *both* polarities of the *primary* remanence of the intrusive are recorded in the fine-grained marginal phase, while in the coarse-grained phase only one polarity is observed, the same as that in the baked sediments. In contrast, it is the overprint (OV1) in the coarse-grained phase that has dual polarity, while the overprint (OV2) remanence in the fine-grained phase has only single polarity. A simple thermoremanent magnetization (TRM) model of remanence acquisition appears not to be applicable. Further, the large differences in magnetic properties and remanence directions of the two overprints (OV1 and OV2) suggest they were probably acquired by different mechanisms. The low T_{UB} (unblocking temperature) overprint (OV2) of the fine-grained phase closely approximates to the present Earth's field direction at the sampling locality, and also to a regional metamorphic magnetization recognized by Morris (1977) in the adjacent Gowganda tillites. On the basis of the available evidence we cannot discern which explanation is applicable. The overprint remanence of the coarse-grained phase (OV1) is a distinct dual-polarity remanence which represents a substantial thermo-chemical resetting of the local remanence system. The timing of these overprint events relative to one another and to the time of intrusion is uncertain. Finally, we note that both dual-polarity remanences have individual components with overlapping blocking temperature spectra. It is beyond the scope of this note to discuss the origin of these magnetizations, but it seems that variations in the extent of hydrothermal alteration giving rise to a new generation of exsolution lamellae could pro-

vide the most viable mechanism (Ade-Hall *et al.* 1971; Ryall and Ade-Hall 1975; Stephenson 1976).

Conclusions

A fully-executed positive contact test between Gowganda argillites and Nipissing diabase shows that the normally magnetized N1 Nipissing direction records a separate phase of diabase intrusion. Symons' (1967) contact test from the Silverfields Mine area on the reversely magnetized N1 Nipissing direction, suggests that this phase of Nipissing diabase intrusion extended over the period of at least one field reversal. Positive contact tests on N2 directed Abitibi and Nipissing (Irving and Naldrett 1977; Roy and Lapointe 1976) conclusively show that there are two quite distinct periods of intrusion. How then does one resolve Symons and Londry's (1975) apparent observation of N1 and N2 directions in the *same* sill and consequent rejection of two intrusive events? As noted earlier the N2 directions exhibit much less dispersion than the N1 directions. Yet, the apparent N2 direction observed by Symons and Londry (1975) is statistically significantly different to all other N2 directions. In fact, they suggest the direction "is a less stable secondary viscous remanent magnetization ... acquired during Recent times." This seems unlikely for two reasons. Firstly, many of the specimens retaining the apparent N2 direction have median destructive fields similar to, and in excess of, those with the N1 direction. Secondly, the apparent N2 and N1 remanences are found to occur only in quite distinct areas of the sill. The zone between the two regions is marked by magnetizations that have poorly grouped remanence directions (sites 4, 11, 14, 15, 16, 17, 20, 23). One possible interpretation of this feature is that the apparent N2 direction has either been partially, or totally overprinted on the N1 remanence. Regional metamorphism during the Penokean Orogeny offers an attractive method by which this overprinting may be effected. During this orogeny metamorphism of the Gowganda argillites gave rise to a distinct overprint direction (Gowganda-B, Morris (1977)). The apparent N2 direction of Symons and Londry is statistically identical to the Gowganda-B direction. Thus, a Penokean metamorphism explanation of this apparent N2 is favoured.

Quite conclusively then, we can show that there were two distinct phases of Nipissing and Abitibi diabase. As yet we are not certain of their sequence of intrusion. However, if we follow Symons' (1971) evidence from the crosscutting dike relationships, this means that the progressively younger magnetic age sequence should be N1→N2→Gowganda-B. This defines a pole path which trends from south to north, in direct contrast with the now commonly accepted north to south trending pole path Track 5 (Roy and Lapointe 1976; Irving and McGlynn 1976).

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